BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MALAKAND

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

PHYSICS

Past Paper 2025 • Paper Code: 2

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** 2

Note: There are Three Sections in this paper i.e. A, B and C. Attempt Section-A and return it to the Superintendent within the given time. No marks will be awarded for Cutting, Erasing and Overwriting. Mobile Phone is strictly prohibited in Examination Hall. Q.1. Choose the correct option and shade one of (A, B, C, D) in the given Bubble Answer Sheet.

Q1. PN junction when reversed act as _____.

(A) Capacitor
(B) Inductor
(C) ON Switch
(D) OFF Switch

Q2. When both the junctions are forward biased, it is _____.

(A) Active Region
(B) Cut-off Region
(C) Saturation Region
(D) Q-Point

Q3. Pair production cannot take place in vacuum because _____ is not conserved.

(A) Mass
(B) Momentum
(C) Energy
(D) Charge

Q4. By doubling the absolute temperature of black-body, the total energy radiated will change by a factor of _____.

(A) 0.5
(B) 2
(C) 4
(D) 16

Q5. Which of the following series of hydrogen spectra lies in the visible region of the spectrum?

(A) Lyman
(B) Balmer
(C) Paschen
(D) Brackett

Q6. Electromagnetic force is caused by exchange of _____.

(A) Electrons
(B) Protons
(C) Neutrons
(D) Photons

Q7. One curie is equal to _____

(A) 3.70 × 10^-10 Bq
(B) 3.70 × 10^10 Bq
(C) 1 × 10^10 Bq
(D) 1MBq

Q8. Which of the following is a boson?

(A) Proton
(B) Neutrino
(C) Photon
(D) Pion

Q9. The electric field intensity due to infinite sheet of charges is given by $\vec{E}$ = _____

(A) $\sigma/\varepsilon_0 \hat{r}$
(B) $\sigma/2\varepsilon_0 \hat{r}$
(C) Q/ε_0
(D) $\delta A/\varepsilon_0$

Q10. 1GeV = _____

(A) 10^9 ev
(B) 10^6 ev
(C) 10^12 ev
(D) 10^15 ev

Q11. The electric field intensity inside a conductor is taken _____

(A) Positive
(B) Negative
(C) Constant
(D) Zero

Q12. The particular temperature at which the thermo emf becomes zero is called _____ temperature.

(A) Neutral
(B) Curie
(C) Inversion
(D) Critical

Q13. How much voltage is required to make 2A through a resistance of 8Ω.

(A) 16AΩ
(B) $16A\Omega^{-1}$
(C) $16\Omega A^{-1}$
(D) $A^{-1}\Omega^{-1}$

Q14. If the charge is at rest in magnetic field, then force on it is _____

(A) qvB
(B) qvB sin θ
(C) qvB cos θ
(D) zero

Q15. When a coil rotate in magnetic field, the induced current in it _____

(A) Continuously changes
(B) Remain same
(C) Becomes maximum
(D) Becomes zero

Q16. The peak value of alternating current is $5\sqrt{2}$. The mean square value of current will be _____.

(A) 5A
(B) 2.5A
(C) $5\sqrt{2}$A
(D) 5^2

Q17. For Young's Modulus, Y = _____

(A) $F/A\theta$
(B) $\Delta P/(-\Delta V/V)$
(C) $Fl/A\Delta L$
(D) F/A

Q18. A wire is stretched to double of its length. Its strain is _____

(A) 0
(B) 0.5
(C) 1
(D) 2

SECTION "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Q2. Give Answers to any TEN (10) questions of the following. All questions carry equal marks.

i. Define: Resistance, Conductance, Resistivity, Conductivity.

ii. How can electric potential be high, when electric potential energy is relatively low?

iii. Show that maximum power output $P_{out} = \varepsilon^2 R/((R-r)^2+4Rr)$

iv. Why voltage is always measured between two points?

v. How does a current carrying coil behave like a bar magnet?

vi. Differentiate between permeability and permittivity.

vii. Show that the relation $\in = \Delta\Phi/\Delta t$ is dimensionally correct.

viii. Show that reactance is measured in ohms for both inductors and capacitors.

ix. How can conductivity of a semi-conductor be raised?

x. If an electron and a proton have the same de-Broglie wavelength, which particle has greater speed?

xi. Why it was necessary to quantize angular momentum in Bohr's model of the atom?

xii. Crystal lattices can be examined with x-rays but not UV. Why?

xiii. What factors make a fusion reaction difficult to achieve?

SECTION "C"

Maximum Marks: 27

Note: Attempt any THREE (03) questions from the following. Each question carries equal marks.

Q3. (Marks 09)

a) [05] State and explain Coulomb's law of electrostatics.

b) [04] Calculate the resistance of a copper wire having length 2km and 22mm^2 cross-sectional area. Assume the resistivity is 18×10^{-9} Ω.m.

Q4. (Marks 09)

a) [05] State and explain Faraday's law of electromagnetic induction. Differentiate between statically and dynamically induced emf.

b) [04] For the same RLC series AC circuit having 40Ω resistor, 3mH inductor and 5μF capacitor. Find the resonant frequency.

Q5. (Marks 09)

a) [05] State Einstein's postulates of special theory of relativity. Discuss its various results.

b) [04] Find the wavelength of electron that has been accelerated through a potential difference of 200V.

Q6. (Marks 09) Give a brief explanation of any two of the following.

i. Gauss's Law

ii. The Uncertainty Principle

iii. Nuclear Fusion